

Audio-driven Talking Face Video Generation with Emotion

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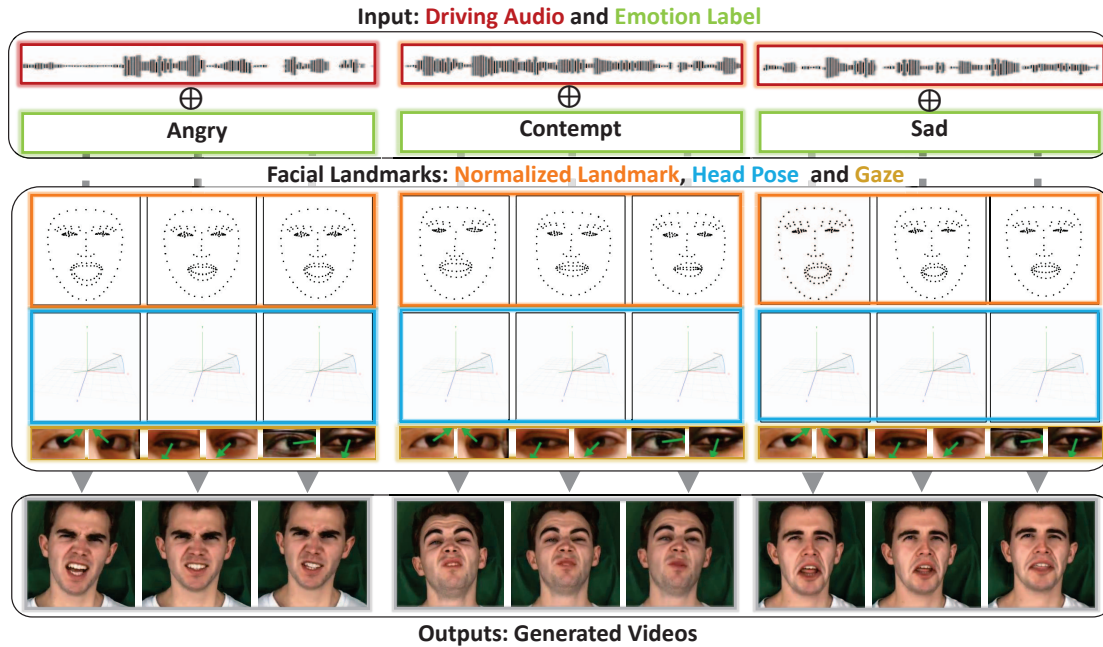


Figure 1: We propose an advanced two-stage framework designed to synthesize talking faces with vivid emotional appearances. In the first stage, our method takes the driving audio and an emotion label as input to synthesize fine-grained facial landmarks, including the normalized landmarks, gaze, and head pose. Based on the generated facial landmarks, we can produce high-quality emotional talking videos in the second stage.

ABSTRACT

Vivid talking face generation has potential applications in virtual reality. Existing methods can generate talking faces that are synchronized with the audio, but typically ignore the accurate expression of emotions. In this paper, we propose an advanced two-step framework to synthesize talking face videos with vivid emotional appearances. The first step is designed to generate emotional fine-grained landmarks, including the normalized landmarks, gaze, and head pose. In the second step, we map the facial landmarks to latent key points, which are then fed into the pre-trained model to generate high-quality face images. Extensive experiments demonstrate the effectiveness of our method.

Index Terms: Digital Human, Talking Face Generation, AIGC

1 INTRODUCTION

The task of talking face generation involves creating a video of a talking face using a still identity image of the speaker and an audio track of their speech content. The primary challenge of this task is to produce realistic facial videos that not only replicate lip

movements but also capture accurate expression of emotions. Existing methods [1, 2, 4] still cannot generate talking faces with vivid emotional expression. In this paper, we propose an advanced speech-driven talking face generation method to synthesize talking faces with vivid emotional appearances. We decompose the task into two sub-tasks: speech-to-landmarks synthesis and landmarks-to-face generation. Given input speech, an emotion label, and an identity image, the proposed speech-to-landmarks module can generate normalized facial landmarks, gaze, and head pose related to emotional categories. The generated gaze and head pose can effectively enhance the accuracy of emotional expression. Subsequently, the proposed landmarks-to-face module builds the mapping from the generated facial landmarks to latent keypoints which are fed into the pre-trained model to complete the generation of high-quality faces. To summarize, our contributions are as follows:

- We proposed an advanced audio-driven talking face generation framework capable of synthesizing talking faces with accurate emotional appearances.
- We introduced an advanced speech-to-landmarks synthesis algorithm which can generate sequences of normalized facial landmarks, gaze, and head pose in an auto-regressive manner.
- Extensive experiments demonstrate the superiority of the proposed method in terms of lip synchronization and the quality of generated faces.

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2 TWO-STAGE TALKING FACE GENERATION

2.1 Speech-to-Landmarks Synthesis.

Facial landmarks are the most critical variables in talking face generation tasks, as their accuracy directly influences the alignment between lip movements and input speech, which is the most crucial metric in this task. Given an input speech, emotion label, and an identity face image, the proposed speech-to-landmarks synthesis is capable of generating sequences of normalized facial landmarks, gaze, and head pose in an auto-regressive manner. We employ Bi-directional Long Short-Term Memory (Bi-LSTM) as the backbone for prediction. Specifically, the network for speech-to-landmarks synthesis consists of three Bi-LSTMs: Landmark LSTM, Gaze LSTM, and Pose LSTM, each performing autoregressive prediction on different facial cue sequences.

Landmark LSTM. We utilize a convolutional neural network (CNN) for encoding audio data, and a multi-layer perceptron (MLP) for encoding 3D facial landmarks. These networks are trained using an L1 loss function, aimed at minimizing the L1 distance between the predicted and normalized ground truth facial landmarks.

Pose LSTM. The natural movement of the head significantly enhances the vividness of the generated video, yet its motion pattern is influenced by the emotional category. Therefore, we have trained an auto-regressive Bi-LSTM to predict the rotation and translation of facial landmarks. The rotation is defined by three angles - yaw, pitch, and roll - while the corresponding translation refers to the displacement of the 3D landmarks along the x-axis, y-axis, and z-axis.

Gaze LSTM. The eyes, pivotal in human interaction, provide substantial information through gaze direction, significantly influencing the emotional appearances of generated videos. The prediction of gaze direction sequences involves a strong dependency on previous gaze patterns. Therefore, we have incorporated a Bi-LSTM model to effectively capture this dependency.

2.2 Landmarks-to-Face Generation.

In the face generation process, we employ 3D facial landmarks created in the speech-to-landmarks synthesis phase, along with input speech and emotion labels, to autoregressively predict latent keypoints within a self-supervised learning framework. Subsequently, using the predicted latent keypoints and the initial facial image, our pre-trained face model produces high-quality facial images through a flow-based warping field as the intermediate variable. Ultimately, this method enables the generation of facial images with a resolution of 256, typically surpassing the quality of previous research.

3 RESULTS

In this set of experiments, we compare our model with state-of-the-art methods for talking head generation on MEAD [4] dataset.

Qualitative Evaluation. Fig.2 presents the qualitative comparison results of our method with other state-of-the-art emotional talking face generation methods. The results clearly demonstrate that our method outperforms other models comprehensively in terms of normalized landmarks, gaze, and head poses. Specifically, our method can generate lip shapes that are more consistent with the input speech, due to the high accuracy of the normalized landmark generation. Furthermore, the natural variations in head pose and gaze in the generated face sequences further demonstrate the effectiveness of the proposed method. Moreover, the faces generated by our method also exhibit a stronger correlation with the input emotion labels. Relevant comparison videos are provided in the supplementary materials.

Quantitative Evaluation. To perform a comprehensive evaluation, we conducted quantitative comparisons between our method and both emotional talking face generation and non-emotional talking face generation approaches. Table 1 reports the quantitative experimental results. Apart from FID, our method outperforms other

Table 1: Quantitative results of different talking face generation models on MEAD dataset.

Method	MLD ↓	FLD ↓	SynNet ↑	SSIM ↑	PSNR ↑	FID ↓
MEAD [4]	2.52	3.16	-	0.68	28.61	22.52
EVP [2]	2.45	3.01	-	0.71	29.53	7.99
EAMM [1]	2.41	2.55	2.26	0.66	29.29	-
EMMN [3]	2.78	2.87	3.57	0.66	29.38	-
Ours	2.08	1.99	4.72	0.74	30.98	8.62

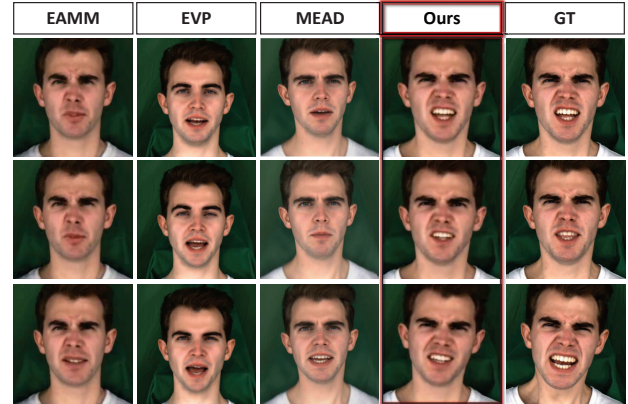


Figure 2: Qualitative comparison of the results generated under 'angry' category with other state-of-the-art models.

approaches significantly in terms of MLD, FLD, and the confidence of SynNet, owing to the joint contributions from all three modules we proposed. The excellent performance of EVP [2] on the FID metric is mainly due to their investment of higher costs in training independent face generation models for each individual. The improvements in MLD, FLD and SynNet shows that our approach can generate more accurate facial landmarks and achieve better lip alignment, which aligns with the qualitative results shown in Fig.2. The results of SSIM, PSNR, and FID also confirm that the proposed method can generate high-quality facial images.

4 CONCLUSION

Emotional talking face generation is a crucial part in digital human which has extensive application in VR. In this paper, we propose a two-stage framework to generate talking face videos with accurate emotional appearances. In the first stage, we generated vivid fine-grained facial landmarks in an auto-regressive manner. In the second stage, we adopted a pre-trained generative model to map generated fine-grained facial landmarks into high-quality facial images. Extensive experiments demonstrate the effectiveness of our model.

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